

Advanced (Carolina Reaper)

- Q1.** In a survey, 25 students like coffee and 15 like tea. The two-way table shows that 10 students like both. What is the total number of students who like at least one of the two drinks?
(A) 30
(B) 35
(C) 40
(D) 45
(E) 50
- Q2.** A two-way table shows the relationship between the type of music and the age group of people. If 60
(A) 0.2
(B) 0.4
(C) 0.6
(D) 0.8
(E) 1.0
- Q3.** In a two-way table, the row total for 'yes' is 25 and the column total for 'agree' is 30. If there are 10 people who said 'yes' and 'agree', what is the $\frac{\text{number of people who said 'yes' and 'agree'}}{\text{total number of people}}$?
(A) $\frac{1}{6}$
(B) $\frac{1}{3}$
(C) $\frac{1}{2}$
(D) $\frac{2}{3}$
(E) $\frac{3}{4}$
- Q4.** undefined
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q5.** A two-way table shows that 30 people like reading and 20 people like writing. If 10 people like both, what is the total number of people who like reading or writing?
(A) 30
(B) 35
(C) 40
(D) 45
(E) 50
- Q6.** In a survey, 50 people said they like sports and 30 said they like music. The two-way table shows that 15 people like both. What is the $\frac{\text{number of people who like both}}{\text{total number of people who like sports}}$?
(A) $\frac{1}{5}$
(B) $\frac{1}{3}$
(C) $\frac{1}{2}$
(D) $\frac{2}{3}$
(E) $\frac{3}{4}$
- Q7.** A two-way table shows the relationship between the type of food and the age group of people. If 20
(A) 0.1
(B) 0.2
(C) 0.3
(D) 0.4
(E) 0.5
- Q8.** In a two-way table, the row total for 'yes' is 20 and the column total for 'agree' is 25. If there are 5 people who said 'yes' and 'agree', what is the $\frac{\text{number of people who said 'yes' and 'agree'}}{\text{total number of people}}$?
(A) $\frac{1}{10}$
(B) $\frac{1}{5}$
(C) $\frac{1}{4}$
(D) $\frac{1}{3}$
(E) $\frac{1}{2}$

Open Ended Questions (FRQ)

- Q9.** A survey was conducted to determine the relationship between the type of transportation used to commute to work and the level of education. The two-way table shows the results. Calculate the relative frequency of people who use a car to commute to work and have a bachelor's degree.
- Q10.** A company conducted a survey to determine the relationship between the type of product purchased and the age group of the customer. The two-way table shows the results. Analyze the association between the type of product and the age group of the customer. Show all your work and explain your answer.

Chef's Tips

- Read each question carefully and choose the correct answer.
- Use the space provided for free response questions.
- Show all your work and explain your answer for free response questions.